

Midterm Summary · Sessions 1 to 6

MATH60033A · Quantitative Methods in International Business

Thierry Warin, PhD · HEC Montréal · A2026

How to use these six pages. A map, not a substitute for the sessions: for each one, the ideas to *state*, the formulas to *use* with a calculator, and a worked example with its own numbers. The last page lists what loses marks. Then sit the sample paper (`assessments/midterm/EXAM.md`) under exam conditions, and only afterwards open `SOLUTIONS.md`.

The paper. Three hours, on paper, in pen. A simple calculator; no notes, no computer, no internet. Four parts: **A** definitions (state it precisely) · **B** calculations (derive and compute, working shown) · **C** diagnosis (read an output, find the planted fault) · **D** judgement (say what a number does and does not license). Working is marked: a clean method with an arithmetic slip earns most of the marks. The midterm examines **Sessions 1 to 6**, all covered here. Session 7, the replication workshop, reviews them on the five papers' own data.

Session 1 · What the course asks of you

The course is about **claims**: what a number lets you say, and what it does not. Three habits run through every answer. **Every number has units**: “one more point of X is associated with 0.31 more units of Y, holding Z fixed”, not “a slope of 0.31”. **Association is not causation**: unless the design isolates a cause, write *is associated with*, never *drives* or *causes*. **Say what you would check**: name the diagnostic that could prove you wrong.

Session 2 · Describing a variable before you model it

Centre

- **Mean** $\bar{x} = \frac{1}{n} \sum_i x_i$: the balance point. Pulled by extreme values.
- **Median**: the middle observation. Robust to outliers; answers “what is typical?”.
- **Trimmed mean**: the mean after dropping a fixed share (e.g. 10%) from each tail; a compromise.
- Mean > median suggests a long **right** tail; mean < median a long **left** tail. For a skewed variable, report the median (or both) and say why.

Spread

- **Sample variance** $s^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2$. Divide by $n - 1$ because $\bar{x}$ was estimated from the same data: the deviations sum to zero, so only $n - 1$ are free. Dividing by n would underestimate the variance.
- **Standard deviation** $s = \sqrt{s^2}$, in the units of x . **IQR** = $Q_3 - Q_1$, the spread of the middle half, robust to outliers.
- **Standard error** of the mean: $SE(\bar{x}) = s/\sqrt{n}$. The SD is the spread of the **observations**; the SE is the spread of a **number computed from them**. More data shrinks the SE, not the SD.

Shape

$$g_1 = \frac{1}{n} \frac{\sum_i (x_i - \bar{x})^3}{s^3} \qquad g_2 = \frac{1}{n} \frac{\sum_i (x_i - \bar{x})^4}{s^4} - 3$$

- $g_1 > 0$: right-skewed; $g_1 < 0$: left-skewed. $g_2 > 0$: heavier tails than a normal (**excess kurtosis**; a normal has $g_2 = 0$).
- **Tests**. Substantially skewed if $|g_1| > 2\sqrt{6/n}$; substantially kurtic if $|g_2| > 4\sqrt{6/n}$. Always report the statistic **with** its threshold.
- **Empirical rule** (68–95–99.7% within 1, 2, 3 SDs) holds only for a roughly **normal** variable. Symmetric is not the same as normal, and the rule does not apply to a variable you have shown to be skewed.

Worked example. $n = 24$, $\bar{x} = 50$, median = 46, $s = 10$, $\sum (x_i - \bar{x})^3 = 26,400$, $\sum (x_i - \bar{x})^4 = 1,020,000$.

$g_1 = \frac{1}{24} \cdot \frac{26,400}{1,000} = 1.10$; threshold $2\sqrt{6/24} = 2(0.5) = 1.00 \rightarrow$ substantially **right**-skewed, consistent with mean > median.

$g_2 = \frac{1}{24} \cdot \frac{1,020,000}{10,000} - 3 = 4.25 - 3 = 1.25$; threshold $4(0.5) = 2.00 \rightarrow$ not substantially kurtic. Report the **median** as the centre.

Session 3 · Can the regression be trusted?

The model and the fit

$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \varepsilon_i$. **Least squares** picks the $\hat{\beta}$ that minimise $\sum_i (y_i - \hat{y}_i)^2$; in matrix form $\hat{\beta} = (X^\top X)^{-1} X^\top y$. The residual is $e_i = y_i - \hat{y}_i$.

- **Slope in units:** a one-unit rise in x_j is associated with a $\hat{\beta}_j$ -unit change in y , **holding the other regressors fixed**.
- **What “controlling for” means (Frisch–Waugh–Lovell).** $\hat{\beta}_j$ equals the slope from regressing the part of y not explained by the other regressors on the part of x_j not explained by them. It is a **partial** relationship, not a marginal one.
- **Omitted variable bias.** If the true model includes z with coefficient β_z , and $z = \delta_0 + \delta_1 x + v$, then the short regression of y on x alone gives $\mathbb{E}[\hat{\beta}_1^{short}] = \beta_1 + \beta_z \delta_1$. The bias has the sign of $\beta_z \times \delta_1$.
- **Unbiasedness** needs linearity, $\mathbb{E}[\varepsilon | X] = 0$ (exogeneity) and full column rank. Constant variance is **not** needed for unbiasedness, only for the usual standard errors and for efficiency.

Adequacy: how much does the line explain?

$R^2 = 1 - \frac{SSR}{SST}$, the share of the variance of y explained. **Adjusted R^2** $= 1 - (1 - R^2) \frac{n-1}{n-k-1}$ penalises extra regressors. **RSE** $= \sqrt{SSR/(n-k-1)}$, the typical size of a residual, in units of y . A high R^2 says nothing about causation or about whether the assumptions hold.

Inference

$t = \hat{\beta}_j / SE(\hat{\beta}_j)$; 95% interval $\hat{\beta}_j \pm 1.96 SE(\hat{\beta}_j)$. Significant at 5% when $|t| > 1.96$, i.e. when the interval excludes 0. **Not significant is not “no effect”**: a wide interval that includes both 0 and large values is simply uninformative.

Validity: do the assumptions hold?

The **residuals-versus-fitted** plot is the instrument. Look for:

- **Linearity** — a curve in the residuals means the functional form is wrong (add a quadratic, transform).
- **Nearly normal residuals** — a Q-Q plot; matters most for intervals in small samples.
- **Constant variance** — a **fan** (residuals spread out as fitted values grow; clearer on a **scale–location** plot) is heteroskedasticity. The coefficients stay unbiased but the classical SEs are wrong $\rightarrow$ use **robust** (HC) standard errors.
- **Multicollinearity** — a variance inflation factor $VIF_j = 1/(1 - R_j^2)$ above about 10 means x_j is nearly a combination of the others: large SEs, unstable signs.

Robustness: which observations move the answer?

- **Outlier:** a large residual. **Leverage point:** an unusual x . **Influential point:** one whose removal changes the fit — typically high leverage *and* a large residual.
- **Leverage** h_{ii} is the i -th diagonal of the hat matrix $H = X(X^\top X)^{-1} X^\top$; $0 \leq h_{ii} \leq 1$ and $\sum_i h_{ii} = p$, the number of parameters. Average leverage is p/n .
- **Cook’s distance** combines leverage and residual into one number per observation; inspect the spikes (a common flag is $D_i > 4/n$). Investigate an influential point; never delete it silently.

Parsimony: comparing candidate models

$AIC = -2\ell + 2p$ and $BIC = -2\ell + p \ln n$, where ℓ is the maximised log-likelihood. **Lower is better**, and only on the **same rows**. BIC penalises extra parameters more heavily, so the two can disagree. Comparing a few candidates you chose is honest; letting a search pick the model is not model selection (Session 5 shows why).

Worked example. $\hat{\beta}_1 = 0.30$, $SE = 0.12$: $t = 2.5$, interval $0.30 \pm 0.235 = (0.065, 0.535)$, significant at 5%. With standard errors clustered by country the SE becomes 0.20: $t = 1.5$, interval $(-0.092, 0.692)$, no longer significant. The coefficient did not change; what you may claim did.

OVB: $\beta_z = 0.05$, $\delta_1 = 0.4 \rightarrow$ bias $= +0.02$. If the short slope is 0.09, the implied $\beta_1 = 0.07$.

Why not OLS on a 0/1 outcome?

The **linear probability model** $P(Y = 1) = \beta_0 + \beta_1 x$ can predict probabilities below 0 or above 1, worst at extreme x . The **logistic** function keeps them in $(0, 1)$:

$$p(x) = P(Y = 1 | x) = \frac{1}{1 + e^{-x^\top \beta}}, \quad \ln \frac{p}{1-p} = x^\top \beta.$$

- β_j is a change in the **log-odds**. e^{β_j} is an **odds ratio**: the odds are multiplied by e^{β_j} per unit of x_j . An OR of 1.30 means odds 30% higher, **not** a probability 30% higher.
- **Marginal effect** on the probability: $\partial p / \partial x_j = p(1-p)\beta_j$. It depends on where you evaluate it, and is largest at $p = 0.5$.
- Estimated by **maximum likelihood**. **Wald test**: $z = \hat{\beta}_j / SE$, one coefficient at a time. **Likelihood-ratio test**: $LR = 2(\ell_{full} - \ell_{restricted}) \sim \chi_q^2$, for q coefficients jointly (e.g. all the dummies of one categorical variable).
- **McFadden's pseudo- R^2** $= 1 - \ell_{model} / \ell_{null}$. Not a share of variance; values of 0.2–0.4 already indicate a good fit.

Ordinal logit (ordered categories)

$\ln \frac{P(Y \leq j)}{P(Y > j)} = c_j - x^\top \beta$, with $J - 1$ **cutpoints** $c_1 < c_2 < \dots$ and **one** β per predictor. Positive β pushes up the scale. Category probabilities are **differences** of cumulative ones: $P(Y = j) = F(c_j - \eta) - F(c_{j-1} - \eta)$. **Proportional odds**: the same odds ratio applies at every cut — check it by fitting a binary logit at each cut.

Multinomial logit (unordered categories)

Pick a **baseline** category. Each other category k gets its own equation $\ln \frac{P(k)}{P(\text{base})} = x^\top \beta_k$. e^{β_k} is a **relative risk ratio** against the baseline. Probabilities: $P(k) = e^{\eta_k} / \sum_m e^{\eta_m}$, with $\eta_{\text{base}} = 0$. A negative coefficient does **not** mean that category's probability falls: every equation moves at once.

From a probability to a decision

- A threshold turns p into a yes/no. If a missed positive costs c_{FN} and a false alarm c_{FP} , the cost-minimising threshold is $\tau^* = \frac{c_{FP}}{c_{FP} + c_{FN}}$ — not 0.5.
- **Accuracy** is meaningless with a rare outcome: with a 3% base rate, “never” is 97% accurate. Report the base rate, **precision** $= TP / (TP + FP)$ and **recall** $= TP / (TP + FN)$.
- **Discrimination** (AUC: does the model rank positives above negatives?) is not **calibration** (does $p = 0.2$ happen 20% of the time?). A high AUC does not make the probabilities usable.

Worked example. $\hat{\beta} = (-2.0, 0.03, 0.5)$ for (intercept, income in thousands, collateral). Applicant: income 50, collateral 1. $x^\top \hat{\beta} = -2.0 + 0.03(50) + 0.5 = 0 \rightarrow p = 1 / (1 + e^0) = 0.50$. Odds ratio for collateral $e^{0.5} \approx 1.65$: odds 65% higher. Marginal effect of income: $0.5 \times 0.5 \times 0.03 = 0.0075$, i.e. **0.75 percentage points** per thousand. If a missed default costs 3 times a false alarm: $\tau^* = 1 / (1 + 3) = 0.25$; since $0.50 > 0.25$, flag the applicant.

Session 5 · Regularisation

Why a biased estimator can win

Test error = bias² + variance + irreducible noise. OLS is unbiased, but with few rows and many correlated predictors its **variance** is enormous. Accepting a little bias for a large cut in variance lowers the total. The penalty λ is the dial: $\lambda = 0$ is OLS; $\lambda \rightarrow \infty$ sets every coefficient to 0 (predict the mean).

Stepwise selection makes one in-or-out decision at a time; resample the rows and it picks a different model, noise included. “The data chose these variables” is not a finding.

Ridge

$$\hat{\beta}^{ridge} = \arg \min_{\beta} \sum_i (y_i - x_i^\top \beta)^2 + \lambda \sum_j \beta_j^2 = (X^\top X + \lambda I)^{-1} X^\top y$$

Shrinks every coefficient toward 0, never exactly to 0; shrinks most where predictors are nearly collinear. $X^\top X + \lambda I$ is invertible for any $\lambda > 0$, **even when** $p > n$. **Standardise first**: the penalty is scale-dependent.

Lasso and soft-thresholding

The lasso penalises $\lambda \sum_j |\beta_j|$. Its diamond-shaped budget has **corners on the axes**, so some coefficients are **exactly 0**: it selects. Coordinate update (standardised columns, ρ_j = covariance of x_j with the partial residual):

$$\hat{\beta}_j = S_\lambda(\rho_j) = \text{sign}(\rho_j) (|\rho_j| - \lambda)_+ \quad (\text{zero if } |\rho_j| \leq \lambda; \text{ else shrink by } \lambda)$$

Limits to state: with $p > n$ it keeps at most n variables; among correlated predictors it keeps one, almost arbitrarily; the selected set is not “the true model”.

Elastic net

Mixes both penalties with weight α ($\alpha = 1$ lasso, $\alpha = 0$ ridge). Update: $\hat{\beta}_j = S_{\lambda\alpha}(\rho_j) / (1 + \lambda(1 - \alpha))$. The **grouping effect**: correlated predictors enter together rather than one standing in for all.

Choosing λ , and what not to claim

- **K -fold cross-validation**: fit on $K - 1$ folds, score on the held-out fold, repeat, average; keep the best λ . Larger $K \rightarrow$ less bias (each fit uses more data) but more variance and more computation; $K = 5$ or 10 is the compromise. λ_{1se} = the largest λ within one SE of the best: a smaller model at no visible cost.
- **Leakage**: imputation, standardisation and feature screening must happen **inside** each fold; for time series, split by time, never shuffle. Otherwise the CV score is optimistic.
- **p-values after selection are not valid**: the same data chose the variables and then tested them.

Worked example — ridge. $X^\top X = \begin{pmatrix} 5 & 2 \\ 2 & 4 \end{pmatrix}$, $X^\top y = \begin{pmatrix} 9 \\ 8 \end{pmatrix}$. OLS: $\det = 20 - 4 = 16$, inverse $\frac{1}{16} \begin{pmatrix} 4 & -2 \\ -2 & 5 \end{pmatrix}$, $\hat{\beta} = \frac{1}{16}(20, 22) = (1.250, 1.375)$. Ridge, $\lambda = 1$: $\begin{pmatrix} 6 & 2 \\ 2 & 5 \end{pmatrix}$, $\det = 26$, $\hat{\beta} = \frac{1}{26}(29, 30) = (1.115, 1.154)$ — shrinkage 10.8% and 16.1%.

Worked example — soft-thresholding. $\rho = (0.50, -0.12, 0.25)$, $\lambda = 0.15$. Lasso: $(0.35, 0, 0.10)$ — two survive. Elastic net, $\alpha = 0.5$: threshold 0.075, divide by 1.075 $\rightarrow (0.395, -0.042, 0.163)$ — all three survive.

Panel data

A **panel** follows the same units (countries, firms) over time. Observations of the same country are not independent, so **pooled OLS with classical standard errors understates uncertainty**. Use standard errors **clustered by unit**.

- **Fixed effects (FE)**: one intercept per unit. Uses only variation *within* each unit over time, so it removes every time-invariant characteristic (geography, institutions, culture) — at the cost of not estimating the effect of anything that does not vary over time.
- **Random effects (RE)**: treats unit intercepts as random draws, **assuming they are uncorrelated with the regressors**. More efficient; biased if that assumption fails.
- **Hausman test**: compares FE and RE. A significant difference says the RE assumption fails → prefer FE. Choose on the assumption, **never** on which one gives the significant answer.

Dummies, interactions and curvature

- A categorical variable with m levels enters as $m - 1$ **dummies**; each coefficient is a difference from the **reference** level, which you must name.
- **Interaction** $y = \beta_0 + \beta_1 x + \beta_2 d + \beta_3 (x \times d)$: the slope of x is β_1 when $d = 0$ and $\beta_1 + \beta_3$ when $d = 1$. Report the slope **in each group**, not β_3 alone.
- **Quadratic** $y = \beta_0 + \beta_1 x + \beta_2 x^2$: the effect of x is $\beta_1 + 2\beta_2 x$ and depends on x ; the turning point is $x^* = -\beta_1 / (2\beta_2)$. Never interpret β_1 on its own.
- **Comparing models**: F-test or LR test for nested models; AIC/BIC for non-nested ones, always on the same rows.

Worked example. $\hat{y} = 1.0 + 0.8x - 0.02x^2$: effect of x at $x = 10$ is $0.8 - 0.04(10) = 0.40$; turning point $x^* = 0.8/0.04 = 20$. With $\hat{y} = 2 + 0.5x + 1.2d - 0.3(x \times d)$: slope 0.5 when $d = 0$, 0.2 when $d = 1$.

What loses marks, session by session

- 2 · a summary pasted without interpretation · a skewness or kurtosis without its threshold · calling a variable normal because it is symmetric · the empirical rule on a skewed variable.
- 3 · a slope without units or without "holding the others fixed" · classical SEs under a visible fan · reading "not significant" as "no effect" · deleting an influential point silently · comparing AIC on different rows.
- 4 · log-odds reported as probabilities · an odds ratio read as a relative risk · accuracy on a rare outcome with no base rate · a 0.5 threshold with unequal costs.
- 5 · penalising unstandardised predictors · standardising or screening before the CV split · the lasso's selected set called "the variables that matter" · post-selection p-values · a dropped variable read as "no effect".
- 6 · pooled OLS on a panel with classical SEs · FE or RE chosen by significance · an interaction without the slope in each group · a quadratic read from β_1 alone.

Writing a Part D answer

Part D rewards a **well-argued refusal**. A strong answer does four things, in order:

1. **Say what the number is.** "Seven variables that jointly predict well at one λ , on one sample" — not "the drivers".
2. **Say what it does not license**, and why: no causal design, a constructed measure, selection that invalidates the p-values.
3. **Name the evidence that would change your answer:** a design that isolates the cause, a validation of the measure, a stability check across resamples, an out-of-sample test.
4. **Recommend what to do meanwhile:** report the association with its uncertainty, run the named diagnostic first, and hold any decision that assumes causation.

Formulas worth carrying in your head. s^2 with $n-1$ · $SE = s/\sqrt{n}$ · $2\sqrt{6/n}$ and $4\sqrt{6/n}$ · $\hat{\beta} = (X^\top X)^{-1}X^\top y$ · $t = \hat{\beta}/SE$ and $\hat{\beta} \pm 1.96 SE$ · $OVB = \beta_z \delta_1$ · $\sum h_{ii} = p$ · $AIC = -2\ell + 2p$ · $p = 1/(1 + e^{-x^\top \beta})$ · odds ratio e^β · $p(1-p)\beta$ · $LR = 2\Delta\ell$ · $\tau^* = c_{FP}/(c_{FP} + c_{FN})$ · $(X^\top X + \lambda I)^{-1}X^\top y$ · $S_\lambda(\rho)$ · $S_{\lambda\alpha}(\rho)/(1 + \lambda(1 - \alpha))$ · $\beta_1 + 2\beta_2 x$ and $-\beta_1/(2\beta_2)$.